

Data Analytics (DA)

Module -2

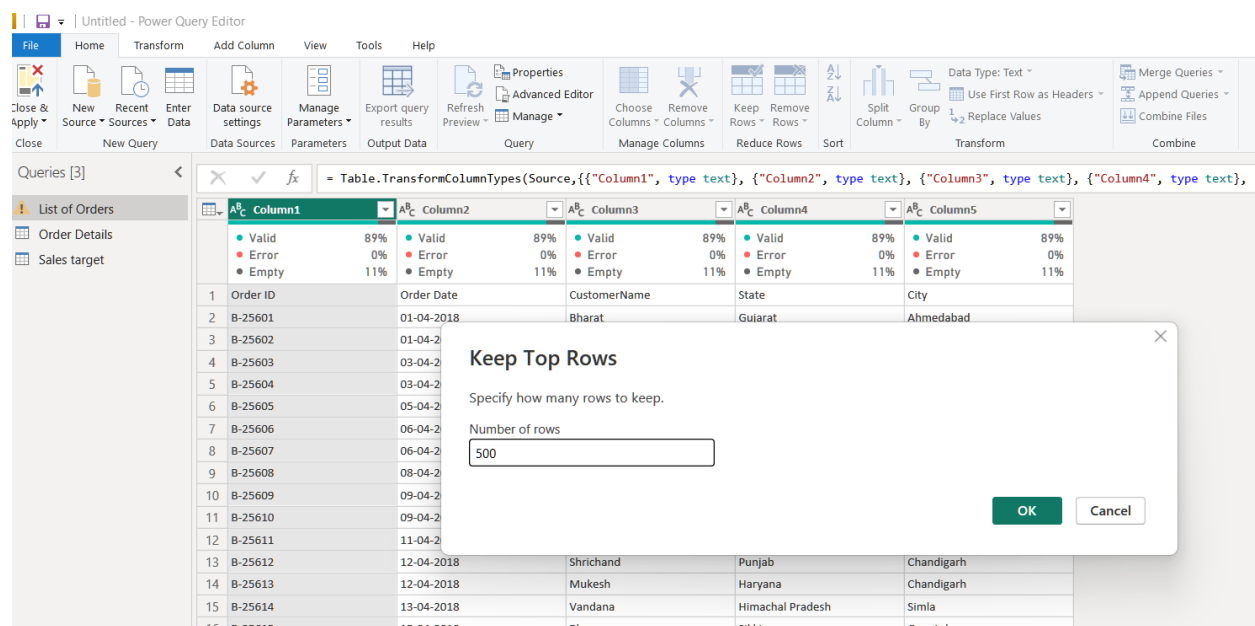
Power BI Assignment 1 – Data Transformation & Data Modeling

Data Import & Transformation:

Imported List of Orders.csv, Order Details.csv, and Sales Target.csv into Power BI using the Get Data option. Opened all datasets in Power Query Editor through the Transform Data feature for further data cleaning and transformation.

Row Limiting & Data Types:

Restricted the List of Orders dataset to the first 500 rows using the Keep Top Rows option in Power Query Editor. Converted the Order Date column to Date data type and changed the Amount and Target columns to Fixed Decimal Number format for accurate analysis and calculations.



Power Query Editor interface showing the 'Keep Top Rows' dialog box. The background table has the following data:

Column1	Column2	Column3	Column4	Column5
Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2			
B-25603	03-04-2			
B-25604	03-04-2			
B-25605	05-04-2			
B-25606	06-04-2			
B-25607	06-04-2			
B-25608	08-04-2			
B-25609	09-04-2			
B-25610	09-04-2			
B-25611	11-04-2			
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Rhuma	Chhittim	Ganetnk

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply * New Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Columns * Keep Remove Rows * Rows * Sort Split Column * Group By Data Type: Text * Use First Row as Headers * Merge Queries * Append Queries * Combine Files * Combine

Queries [3] - Table.FirstN(#"Changed Type",500)

List of Orders Order Details Sales target

Column1	Column2	Column3	Column4	Column5
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
Order ID	Order Date	CustomerName	State	City
1 B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2 B-25602	01-04-2018	Pearl	Maharashtra	Pune
4 B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
5 B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
6 B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
7 B-25606	06-04-2018	Hazel	Karnataka	Bangalore
8 B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
9 B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
10 B-25609	09-04-2018	Itesh	Uttar Pradesh	Lucknow
11 B-25610	09-04-2018	Yogesh	Bihar	Patna
12 B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
13 B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
14 B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
15 B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
16 B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
17 B-25616	15-04-2018	Kanak	Goa	Goa
18 B-25617	17-04-2018	Sagar	Nagaland	Kohima
19 B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
20 B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
21 B-25620	20-04-2018	Sarita	Maharashtra	Pune
22 B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
23 B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
24 B-25623	22-04-2018	Atharv	West Bengal	Kolkata
25 B-25624	22-04-2018	Vini	Karnataka	Bangalore

5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

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Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply * New Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Columns * Keep Remove Rows * Rows * Sort Split Column * Group By Data Type: Date * Use First Row as Headers * Merge Queries * Append Queries * Combine Files * Combine

Queries [3] - Table.TransformColumnTypes(#"Kept First Rows", {"(\"Order Date\", type date)}, \"en-GB\")

List of Orders Order Details Sales target

Order ID	Order Date	CustomerName	State	City
1 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad
2 B-25602	4/1/2018	Pearl	Maharashtra	Pune
3 B-25603	4/3/2018	Jahan	Madhya Pradesh	Bhopal
4 B-25604	4/3/2018	Divsha	Rajasthan	Jaipur
5 B-25605	4/5/2018	Kasheen	West Bengal	Kolkata
6 B-25606	4/6/2018	Hazel	Karnataka	Bangalore
7 B-25607	4/6/2018	Sonakshi	Jammu and Kashmir	Kashmir
8 B-25608	4/8/2018	Aarushi	Tamil Nadu	Chennai
9 B-25609	4/9/2018	Itesh	Uttar Pradesh	Lucknow
10 B-25610	4/9/2018	Yogesh	Bihar	Patna
11 B-25611	4/11/2018	Anita	Kerala	Thiruvananthapuram
12 B-25612	4/12/2018	Shrichand	Punjab	Chandigarh
13 B-25613	4/12/2018	Mukesh	Haryana	Chandigarh
14 B-25614	4/13/2018	Vandana	Himachal Pradesh	Simla
15 B-25615	4/15/2018	Bhavna	Sikkim	Gangtok
16 B-25616	4/15/2018	Kanak	Goa	Goa
17 B-25617	4/17/2018	Sagar	Nagaland	Kohima
18 B-25618	4/18/2018	Manju	Andhra Pradesh	Hyderabad
19 B-25619	4/18/2018	Ramesh	Gujarat	Ahmedabad
20 B-25620	4/20/2018	Sarita	Maharashtra	Pune
21 B-25621	4/20/2018	Deepak	Madhya Pradesh	Bhopal
22 B-25622	4/22/2018	Monisha	Rajasthan	Jaipur
23 B-25623	4/23/2018	Atharv	West Bengal	Kolkata
24 B-25624	4/22/2018	Vini	Karnataka	Bangalore
25 B-25625	4/23/2018	Pinky	Jammu and Kashmir	Kashmir
26 B-25626	4/23/2018	Bhishm	Maharashtra	Mumbai
27 B-25627	4/23/2018	Hittika	Madhya Pradesh	Indore
28 B-25628	4/24/2018	Pooja	Bihar	Patna

5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

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Power Query Editor - Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Enter Data * Data source settings * Parameters * Data Sources * Manage Parameters * Export query results * Refresh Preview * Query * Properties * Advanced Editor * Choose Columns * Remove Columns * Keep Rows * Remove Rows * Sort * Split Column * Group By * Data Type: Fixed decimal number * Use First Row as Headers * Replace Values * Merge Queries * Append Queries * Combine Files * Combine

Queries [3]

- List of Orders
- Order Details
- Sales target

Table.TransformColumnTypes(#Promoted Headers,{"Target", Currency.Type})

	Month of Order Date	Category	Target
1	Apr-18	Furniture	10,400.00
2	May-18	Furniture	10,500.00
3	Jun-18	Furniture	10,600.00
4	Jul-18	Furniture	10,800.00
5	Aug-18	Furniture	10,900.00
6	Sep-18	Furniture	11,000.00
7	Oct-18	Furniture	11,100.00
8	Nov-18	Furniture	11,300.00
9	Dec-18	Furniture	11,400.00
10	Jan-19	Furniture	11,500.00
11	Feb-19	Furniture	11,600.00
12	Mar-19	Furniture	11,800.00
13	Apr-18	Clothing	12,000.00
14	May-18	Clothing	12,000.00
15	Jun-18	Clothing	12,000.00
16	Jul-18	Clothing	14,000.00
17	Aug-18	Clothing	14,000.00
18	Sep-18	Clothing	14,000.00
19	Oct-18	Clothing	16,000.00
20	Nov-18	Clothing	16,000.00
21	Dec-18	Clothing	16,000.00
22	Jan-19	Clothing	16,000.00
23	Feb-19	Clothing	16,000.00
24	Mar-19	Clothing	16,000.00
25	Apr-18	Electronics	9,000.00
26	May-18	Electronics	9,000.00
27	Jun-18	Electronics	9,000.00
28	Jul-18	Electronics	9,000.00

Query Settings

PROPERTIES

Name

Sales target

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type

Power Query Editor - Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Enter Data * Data source settings * Parameters * Data Sources * Manage Parameters * Export query results * Refresh Preview * Query * Properties * Advanced Editor * Choose Columns * Remove Columns * Keep Rows * Remove Rows * Sort * Split Column * Group By * Data Type: Fixed decimal number * Use First Row as Headers * Replace Values * Merge Queries * Append Queries * Combine Files * Combine

Queries [3]

- List of Orders
- Order Details
- Sales target

Table.TransformColumnTypes(#Promoted Headers,{"Order ID", type text, ("Amount", Currency.Type), ("Profit", Int64.Type), ("Quantity", type text, ("Category", type text, ("Sub-Category", type text))})

	Order ID	Amount	Profit	Quantity	Category	Sub-Category
1	B-25601	1,275.00	-1148	7	Furniture	Bookcases
2	B-25601	66.00	-12	5	Clothing	Stole
3	B-25601	8.00	-2	3	Clothing	Hankerchief
4	B-25601	80.00	-56	4	Electronics	Electronic Games
5	B-25602	168.00	-111	2	Electronics	Phones
6	B-25602	424.00	-272	5	Electronics	Phones
7	B-25602	2,617.00	1151	4	Electronics	Phones
8	B-25602	561.00	212	3	Clothing	Saree
9	B-25602	119.00	-5	8	Clothing	Saree
10	B-25603	1,355.00	-60	5	Clothing	Trousers
11	B-25603	24.00	-30	1	Furniture	Chairs
12	B-25603	193.00	-166	3	Clothing	Saree
13	B-25603	180.00	5	3	Clothing	Trousers
14	B-25603	116.00	16	4	Clothing	Stole
15	B-25603	107.00	36	6	Clothing	Stole
16	B-25603	12.00	1	2	Clothing	Hankerchief
17	B-25603	38.00	18	1	Clothing	Kurti
18	B-25604	65.00	17	2	Clothing	T-shirt
19	B-25604	157.00	5	9	Clothing	Saree
20	B-25605	75.00	0	7	Clothing	Saree
21	B-25606	87.00	4	2	Clothing	Shirt
22	B-25607	50.00	15	4	Clothing	Leggings
23	B-25608	1,364.00	-1864	5	Furniture	Tables
24	B-25608	476.00	0	3	Furniture	Chairs
25	B-25608	257.00	23	5	Clothing	Hankerchief
26	B-25608	856.00	385	6	Electronics	Printers
27	B-25609	485.00	29	4	Electronics	Electronic Games
28	B-25609	25.00	-5	4	Clothing	Saree

Query Settings

PROPERTIES

Name

Order Details

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type

6 COLUMNS, 999+ ROWS - Column profiling based on top 1000 rows

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Text Formatting:

Formatted the Customer Name column using the Capitalize Each Word option in Power Query Editor to maintain consistent capitalization and improve data quality.

The screenshot shows the Power Query Editor interface. The 'Format' menu is open, and 'Capitalize Each Word' is selected. The main data table is visible, showing columns for Order ID, Order Date, and Customer Name. The Customer Name column is being formatted. The 'APPLIED STEPS' list on the right includes 'Capitalized Each Word'.

Order ID	Order Date	Customer Name	City
1	4/1/2018	Bharat	Ahmedabad
2	4/1/2018	Pearl	Pune
3	4/3/2018	Jahan	Bhopal
4	4/3/2018	Divsha	Jaipur
5	4/5/2018	Kasheen	Kolkata
6	4/6/2018	Hazel	Bangalore
7	4/6/2018	Sonakshi	Kashmir
8	4/8/2018	Aarushi	Chennai
9	4/9/2018	Jitesh	Lucknow
10	4/9/2018	Yogesh	Patna
11	4/11/2018	Anita	Thiruvananthapuram
12	4/12/2018	Shrichand	Chandigarh
13	4/12/2018	Mukesh	Chandigarh
14	4/13/2018	Vandana	Simla
15	4/15/2018	Bhavna	Gangtok
16	4/15/2018	Kanak	Goa
17	4/17/2018	Sagar	Kohima
18	4/18/2018	Manju	Hyderabad
19	4/18/2018	Ramesh	Ahmedabad
20	4/20/2018	Sarita	Pune
21	4/20/2018	Deepak	Bhopal
22	4/23/2018	Monisha	Jaipur
23	4/23/2018	Atharv	Kolkata
24	4/23/2018	Vini	Bangalore
25	4/23/2018	Pinky	Kashmir
26	4/23/2018	Bhishm	Mumbai
27	4/23/2018	Hittika	Indore
28	4/24/2018	Pooja	Patna

Column Creation:

Merged the City and State columns to create a Location field in the format “City, State”. Added a custom column named Profit Margin using the formula Profit divided by Amount and formatted the result as a percentage for profitability analysis.

Power Query Editor interface showing the 'Merge Columns' dialog box. The dialog is titled 'Merge Columns' and asks to choose how to merge the selected columns. The 'Separator' is set to 'Comma'. The 'New column name (optional)' is set to 'Location'. The background shows a table with columns: Order ID, Order Date, CustomerName, State, and City. The formula bar shows: = Table.TransformColumns(#\"Changed Type with Locale\",{{\"CustomerName\", Text.Proper, type text}}).

Order ID	Order Date	CustomerName	State	City
1	4/1/2018	Bharat	Gujarat	Ahmedabad
2	4/1/2018	Pearl	Maharashtra	Pune
3	4/3/2018	Jahan	Madhya Pradesh	Bhopal
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17	4/1/2018	Sagar	Nagaland	Konima
18	4/18/2018	Manju	Andhra Pradesh	Hyderabad
19	4/18/2018	Ramesh	Gujarat	Ahmedabad
20	4/20/2018	Sarita	Maharashtra	Pune
21	4/20/2018	Deepak	Madhya Pradesh	Bhopal
22	4/22/2018	Monisha	Rajasthan	Jaipur
23	4/22/2018	Atharv	West Bengal	Kolkata
24	4/22/2018	Vini	Karnataka	Bangalore
25	4/23/2018	Pinky	Jammu and Kashmir	Kashmir

Power Query Editor interface showing the 'Merge Columns' dialog box. The dialog is titled 'Merge Columns' and asks to choose how to merge the selected columns. The 'Separator' is set to '--Custom--'. The 'New column name (optional)' is set to 'Location'. The background shows a table with columns: Order ID, Order Date, CustomerName, State, and City. The formula bar shows: = Table.TransformColumns(#\"Changed Type with Locale\",{{\"CustomerName\", Text.Proper, type text}}).

Order ID	Order Date	CustomerName	State	City
1	4/1/2018	Bharat	Gujarat	Ahmedabad
2	4/1/2018	Pearl	Maharashtra	Pune
3	4/3/2018	Jahan	Madhya Pradesh	Bhopal
4				
5				
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15				
16				
17				
18				
19	4/18/2018	Ramesh	Gujarat	Ahmedabad
20	4/20/2018	Sarita	Maharashtra	Pune
21	4/20/2018	Deepak	Madhya Pradesh	Bhopal
22	4/22/2018	Monisha	Rajasthan	Jaipur
23	4/22/2018	Atharv	West Bengal	Kolkata
24	4/22/2018	Vini	Karnataka	Bangalore
25	4/23/2018	Pinky	Jammu and Kashmir	Kashmir

Power Query Editor interface showing a table with columns: Order ID, Order Date, Customer Name, and Location. The formula bar shows: `= Table.CombineColumns(#"Capitalized Each Word",{"State","City"},Combiner.CombineTextByDelimiter(",","QuoteStyle.None"),"Location")`. The table contains 27 rows of data. A red arrow points to the 'Location' column header.

Order ID	Order Date	Customer Name	Location
B-25601	4/1/2018	Bharat	Gujarat, Ahmedabad
B-25602	4/1/2018	Pearl	Maharashtra, Pune
B-25603	4/3/2018	Jahan	Madhya Pradesh, Bhopal
B-25604	4/3/2018	Divsha	Rajasthan, Jaipur
B-25605	4/5/2018	Kasheen	West Bengal, Kolkata
B-25606	4/6/2018	Hazel	Karnataka, Bangalore
B-25607	4/6/2018	Sonakshi	Jammu and Kashmir, Kashmir
B-25608	4/8/2018	Aarushi	Tamil Nadu, Chennai
B-25609	4/9/2018	Jitesh	Uttar Pradesh, Lucknow
B-25610	4/9/2018	Yogesh	Bihar, Patna
B-25611	4/11/2018	Anita	Kerala, Thiruvananthapuram
B-25612	4/12/2018	Shrichand	Punjab, Chandigarh
B-25613	4/12/2018	Mukesh	Haryana, Chandigarh
B-25614	4/13/2018	Vandana	Himachal Pradesh, Simla
B-25615	4/15/2018	Bhavana	Sikkim, Gangtok
B-25616	4/15/2018	Kanak	Goa, Goa
B-25617	4/17/2018	Sagar	Nagaland, Kohima
B-25618	4/18/2018	Manju	Andhra Pradesh, Hyderabad
B-25619	4/18/2018	Ramesh	Gujarat, Ahmedabad
B-25620	4/20/2018	Sarita	Maharashtra, Pune
B-25621	4/20/2018	Deepak	Madhya Pradesh, Bhopal
B-25622	4/22/2018	Monisha	Rajasthan, Jaipur
B-25623	4/22/2018	Atharv	West Bengal, Kolkata
B-25624	4/22/2018	Vini	Karnataka, Bangalore
B-25625	4/23/2018	Pinky	Jammu and Kashmir, Kashmir
B-25626	4/23/2018	Bhishm	Maharashtra, Mumbai
B-25627	4/23/2018	Hitika	Madhya Pradesh, Indore

Power Query Editor interface showing a table with columns: Order ID, Amount, Profit, Quantity, Category, and Sub-Category. The formula bar shows: `= Table.TransformColumnTypes(#"Promoted Headers",{"Order ID", type text}, {"Amount", Currency.Type}, {"Profit", Int64.Type}, {"Quantity",`. A 'Custom Column' dialog box is open, showing the formula: `= [Profit] / [Amount]`. The dialog box also lists available columns: Order ID, Amount, Profit, Quantity, Category, and Sub-Category.

Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25601	87.00	4	2	Clothing	Shirt
B-25602	50.00	15	4	Clothing	Leggings
B-25603	1,364.00	-1864	5	Furniture	Tables
B-25604	476.00	0	3	Furniture	Chairs
B-25605	257.00	23	5	Clothing	Hankchief
B-25606	856.00	385	6	Electronics	Printers
B-25607	485.00	29	4	Electronics	Electronic Games
B-25608	25.00	-5	4	Clothing	Saree

Untitled - Power BI Desktop

File Home Help Table tools Column tools

Name Profit Margin Data type Decimal number Format Percentages Summarization Sum Data category Uncategorized Sort by column Sort Data groups Groups Manage relationships Relationships Calculations New column

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25603	\$12	1	2	Clothing	Hankercchief	8.33%	Profit
B-25608	\$257	23	5	Clothing	Hankercchief	8.95%	Profit
B-25615	\$68	20	5	Clothing	Hankercchief	29.41%	Profit
B-25616	\$42	12	5	Clothing	Hankercchief	28.57%	Profit
B-25624	\$26	12	3	Clothing	Hankercchief	46.15%	Profit
B-25625	\$97	29	2	Clothing	Hankercchief	29.90%	Profit
B-25635	\$40	16	3	Clothing	Hankercchief	40.00%	Profit
B-25638	\$154	39	3	Clothing	Hankercchief	25.32%	Profit
B-25654	\$34	12	3	Clothing	Hankercchief	35.29%	Profit
B-25656	\$6	3	1	Clothing	Hankercchief	50.00%	Profit
B-25656	\$56	18	2	Clothing	Hankercchief	32.14%	Profit
B-25670	\$24	1	2	Clothing	Hankercchief	4.17%	Profit
B-25670	\$14	2	1	Clothing	Hankercchief	14.29%	Profit
B-25730	\$18	8	2	Clothing	Hankercchief	44.44%	Profit
B-25751	\$32	7	3	Clothing	Hankercchief	21.88%	Profit
B-25757	\$106	15	7	Clothing	Hankercchief	14.15%	Profit
B-25757	\$14	5	1	Clothing	Hankercchief	35.71%	Profit
B-25757	\$17	7	3	Clothing	Hankercchief	41.18%	Profit
B-25771	\$148	59	3	Clothing	Hankercchief	39.86%	Profit
B-25783	\$26	2	2	Clothing	Hankercchief	7.69%	Profit
B-25784	\$15	4	1	Clothing	Hankercchief	26.67%	Profit
B-25793	\$18	1	3	Clothing	Hankercchief	5.56%	Profit
B-25797	\$31	1	2	Clothing	Hankercchief	3.23%	Profit
B-25800	\$45	12	7	Clothing	Hankercchief	26.67%	Profit
B-25808	\$270	50	4	Clothing	Hankercchief	23.81%	Profit
B-25809	\$154	54	3	Clothing	Hankercchief	35.06%	Profit
B-25809	\$53	24	1	Clothing	Hankercchief	45.28%	Profit
B-25810	\$36	10	4	Clothing	Hankercchief	27.78%	Profit

Table: Order Details (1,500 rows) Column: Profit Margin (1,155 distinct values)

Table: Order Details (1,500 rows) Column: Profit Margin (1,155 distinct values)

Conditional Column:

Added a conditional column named Profit Status using conditional logic in Power Query Editor. Classified records as Loss, Break-Even, or Profit based on profit values to support profitability analysis.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Jump From Custom Invoke Custom samples - Column Function General

Conditional Column Index Column Duplicate Column Format Extract Statistics Standard Scientific Trigonometry Rounding Information Date Time Duration From Text From Number From Date & Time

Queries [3] List of Orders Order Details Sales target

Table.TransformColumnTypes(#"Added Custom",{"Profit Margin", type number})

Amount Profit Quantity Category Sub-Category Profit Margin

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name: Profit Status

Column Name	Operator	Value	Output
Profit	is less than	0	Loss
Profit	equals	0	Break-Even
Profit	is greater than	0	Profit

OK Cancel

Query Settings

PROPERTIES

Name: Order Details

APPLIED STEPS

Source Promoted Headers Changed Type Added Custom Changed Type1

File Home Transform Add Column View Tools Help

Column From Custom Invoke Custom Examples * Column Function General

Conditional Column Index Column Merge Columns Format Parse Statistics Standard Scientific Trigonometry Rounding Information Date Time Duration From Text From Number From Date & Time

Queries [3]

- List of Orders
- Order Details
- Sales target

fx = Table.AddColumn("#Changed Type1", "Profit Status", each if [Profit] < 0 then "Loss" else if [Profit] = 0 then "Break-Even" else if

	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
1	1,275.00	-1148	7 Furniture	Bookcases	-0.900392157	Loss
2	66.00	-12	5 Clothing	Stole	-0.181818182	Loss
3	8.00	-2	3 Clothing	Hankerchief	-0.25	Loss
4	80.00	-56	4 Electronics	Electronic Games	-0.7	Loss
5	168.00	-111	2 Electronics	Phones	-0.660714286	Loss
6	424.00	-272	5 Electronics	Phones	-0.641509434	Loss
7	2,617.00	1151	4 Electronics	Phones	0.439816584	Profit
8	561.00	212	3 Clothing	Saree	0.377896613	Profit
9	119.00	-5	8 Clothing	Saree	-0.042016807	Loss
10	1,355.00	-60	5 Clothing	Trousers	-0.044280443	Loss
11	24.00	-30	1 Furniture	Chairs	-1.25	Loss
12	193.00	-166	3 Clothing	Saree	-0.860103627	Loss
13	180.00	5	3 Clothing	Trousers	0.027777778	Profit
14	116.00	16	4 Clothing	Stole	0.137931034	Profit
15	107.00	36	6 Clothing	Stole	0.336448598	Profit
16	12.00	1	2 Clothing	Hankerchief	0.083333333	Profit
17	38.00	18	1 Clothing	Kurti	0.473684211	Profit
18	65.00	17	2 Clothing	T-shirt	0.261538462	Profit
19	157.00	5	9 Clothing	Saree	0.031847134	Profit
20	75.00	0	7 Clothing	Saree	0	Break-Even
21	87.00	4	2 Clothing	Shirt	0.045977011	Profit
22	50.00	15	4 Clothing	Leggings	0.3	Profit
23	1,364.00	-1864	5 Furniture	Tables	-1.366568915	Loss
24	476.00	0	3 Furniture	Chairs	0	Break-Even
25	257.00	23	5 Clothing	Hankerchief	0.089494163	Profit
26	856.00	385	6 Electronics	Printers	0.449766355	Profit
27	485.00	29	4 Electronics	Electronic Games	0.059793814	Profit

8 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

Merging Data (Joins):

Merged the List of Orders and Order Details tables using the Order ID field with a Left Outer Join in Power Query Editor. Expanded the required columns and renamed the merged table as Orders Data for unified analysis.

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Enter Data * Data source settings * Manage Parameters * Export query results * Output Data

Queries [3]

- List of Orders
- Order Details
- Sales target

fx = Table.CombineC

Merge

Select a table and matching columns to create a merged table.

List of Orders

Order ID	Order Date	CustomerName	Location
B-25601	4/1/2018	Bharat	Gujarat, Ahmedabad
B-25602	4/1/2018	Pearl	Maharashtra, Pune
B-25603	4/3/2018	Jahan	Madhya Pradesh, Bhopal
B-25604	4/3/2018	Divsha	Rajasthan, Jaipur
B-25605	4/5/2018	Kasheem	West Bengal, Kolkata

Order Details

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25601	1,275.00	-1148	7	Furniture	Bookcases	-0.900392157	Loss
B-25601	66.00	-12	5	Clothing	Stole	-0.181818182	Loss
B-25601	8.00	-2	3	Clothing	Hankerchief	-0.25	Loss
B-25601	80.00	-56	4	Electronics	Electronic Games	-0.7	Loss
B-25602	168.00	-111	2	Electronics	Phones	-0.660714286	Loss

Join Kind
Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

Fuzzy matching options

OK Cancel

Query Settings

Location*)

PROPERTIES

Name
List of Orders

APPLIED STEPS

- Source
- Changed Type
- Promoted Headers
- Changed Type1
- Kept First Rows
- Changed Type with Locale
- Capitalized Each Word
- Merged Columns

4 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 8:39 PM

Power Query Editor interface showing a data table with columns: Order ID, Order Date, CustomerName, Location, Order ID.1, Amount, Profit. The table contains 29 rows of data. The right sidebar shows 'Query Settings' with 'Properties' and 'Applied Steps' sections. The 'Applied Steps' section lists: Source, Changed Type, Promoted Headers, Changed Type1, Kept First Rows, Changed Type with Locale, Capitalized Each Word, Merged Columns, Merged Queries, and Expanded Order Details. The bottom status bar indicates '0 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

Power Query Editor interface showing a data table with columns: Order ID, Order Date, CustomerName, Location, Order ID.1, Amount, Profit. The table contains 29 rows of data. The right sidebar shows 'Query Settings' with 'Properties' and 'Applied Steps' sections. The 'Applied Steps' section lists: Source, Changed Type, Promoted Headers, Changed Type1, Kept First Rows, Changed Type with Locale, Capitalized Each Word, Merged Columns, Merged Queries, and Expanded Order Details. The bottom status bar indicates '0 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

Handling Missing Data & Duplicate Data:

Checked all tables for missing values, nulls, and errors using column quality indicators in Power Query Editor. No missing or erroneous values were found in the final Orders Data table. Duplicate handling was performed carefully using business logic, retaining repeated Order IDs where multiple products belonged to the same order, and removing only completely identical duplicate records.

Power Query Editor interface showing a table with columns: Order ID, Order Date, Customer Name, Location, Amount, Profit, and Quantity. The table contains 1500 rows of data. The 'Applied Steps' pane on the right shows the sequence of transformations, with 'Removed Duplicates' highlighted by a red arrow.

Power Query Editor interface showing a table with columns: Profit, Quantity, Category, Sub-Category, Profit Margin, and Profit Status. The table contains 25 rows of data. The 'Applied Steps' pane on the right shows the sequence of transformations, with 'Removed Duplicates' highlighted by a red arrow.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Enter Data * Data source settings * Data Sources * Manage Parameters * Export query results * Refresh Preview * Properties * Advanced Editor * Choose Columns * Remove Columns * Keep Rows * Remove Rows * Sort * Split Column * Group By * Data Type: Text * Use First Row as Headers * Merge Queries * Append Queries * Combine Files * Combine

Queries [3] - Table.SelectRows(#"Sorted Rows", each ([Location] = "Chennai, Tamil Nadu"))

Orders Data Order Details Sales target

Order ID Order Date CustomerName Location Amount Profit Quantity

1 B-26081 3/22/20 79.00 33

2 B-26081 3/22/20 169.00 0

3 B-26081 3/22/20 359.00 -338

4 B-26081 3/22/20 93.00 -84

5 B-26081 3/22/20 24.00 11

6 B-26081 3/22/20 637.00 50

7 B-26018 2/14/20 61.00 8

8 B-26018 2/14/20 326.00 107

9 B-26008 2/9/20 22.00 4

10 B-26008 2/9/20 206.00 51

11 B-26008 2/9/20 10.00 -1

12 B-26008 2/9/20 57.00 24

13 B-25860 11/15/20 112.00 24

14 B-25788 9/21/20 12.00 3

15 B-25716 7/11/20 58.00 0

16 B-25698 6/23/20 207.00 -153

17 B-25698 6/23/20 27.00 -6

18 B-25698 6/23/20 65.00 -16

19 B-25698 6/23/20 87.00 -83

20 B-25698 6/23/20 7.00 -2

21 B-25698 6/23/20 516.00 -392

22 B-25608 4/8/20 257.00 23

23 B-25608 4/8/20 856.00 385

24 B-25608 4/8/20 476.00 0

25

9 COLUMNS, 25 ROWS Column profiling based on top 1000 rows

OK Cancel

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Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply * New Source * Recent Sources * Enter Data * Data source settings * Data Sources * Manage Parameters * Export query results * Refresh Preview * Properties * Advanced Editor * Choose Columns * Remove Columns * Keep Rows * Remove Rows * Sort * Split Column * Group By * Data Type: Text * Use First Row as Headers * Merge Queries * Append Queries * Combine Files * Combine

Queries [3] - Table.SelectRows(#"Sorted Rows", each ([Location] = "Chennai, Tamil Nadu"))

Orders Data Order Details Sales target

Order ID Order Date CustomerName Location Amount Profit Quantity

1 B-26081 3/22/2019 Aarushi Chennai, Tamil Nadu 79.00 33

2 B-26081 3/22/2019 Aarushi Chennai, Tamil Nadu 169.00 0

3 B-26081 3/22/2019 Aarushi Chennai, Tamil Nadu 359.00 -338

4 B-26081 3/22/2019 Aarushi Chennai, Tamil Nadu 93.00 -84

5 B-26081 3/22/2019 Aarushi Chennai, Tamil Nadu 24.00 11

6 B-26081 3/22/2019 Aarushi Chennai, Tamil Nadu 637.00 50

7 B-26018 2/14/2019 Aarushi Chennai, Tamil Nadu 61.00 8

8 B-26018 2/14/2019 Aarushi Chennai, Tamil Nadu 326.00 107

9 B-26008 2/9/2019 Kalyani Chennai, Tamil Nadu 22.00 4

10 B-26008 2/9/2019 Kalyani Chennai, Tamil Nadu 206.00 51

11 B-26008 2/9/2019 Kalyani Chennai, Tamil Nadu 10.00 -1

12 B-26008 2/9/2019 Kalyani Chennai, Tamil Nadu 57.00 24

13 B-25860 11/15/2018 Akshay Chennai, Tamil Nadu 112.00 24

14 B-25788 9/21/2018 Dinesh Chennai, Tamil Nadu 12.00 3

15 B-25716 7/11/2018 Surabhi Chennai, Tamil Nadu 58.00 0

16 B-25698 6/23/2018 Amisha Chennai, Tamil Nadu 207.00 -153

17 B-25698 6/23/2018 Amisha Chennai, Tamil Nadu 27.00 -6

18 B-25698 6/23/2018 Amisha Chennai, Tamil Nadu 65.00 -16

19 B-25698 6/23/2018 Amisha Chennai, Tamil Nadu 87.00 -83

20 B-25698 6/23/2018 Amisha Chennai, Tamil Nadu 7.00 -2

21 B-25698 6/23/2018 Amisha Chennai, Tamil Nadu 516.00 -392

22 B-25608 4/8/2018 Aarushi Chennai, Tamil Nadu 257.00 23

23 B-25608 4/8/2018 Aarushi Chennai, Tamil Nadu 856.00 385

24 B-25608 4/8/2018 Aarushi Chennai, Tamil Nadu 476.00 0

25

9 COLUMNS, 25 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 8:32

Table.SelectRows(#"Sorted Rows", each true)

MAHA

(Select All Search Results)

☐ Mumbai, Maharashtra

☒ Pune, Maharashtra

List may be incomplete. Load more

OK Cancel

9 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 8:35 A

Table.SelectRows(#"Sorted Rows", each ([Location] = "Pune, Maharashtra"))

MAHA

(Select All Search Results)

☐ Mumbai, Maharashtra

☒ Pune, Maharashtra

List may be incomplete. Load more

OK Cancel

9 COLUMNS, 83 ROWS Column profiling based on top 1000 rows

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Grouping and Aggregating Data:

Duplicated the Order Details and Sales Target tables to perform aggregation operations. Calculated the count of Order IDs, average profit by category, total amount by sub-category, and total sales target by month using the Group By feature in Power Query Editor for summarized business analysis. Duplicated the Sales Target table and aggregated the Target values by Month of Order Date using the Group By transformation in Power Query Editor to generate monthly sales target summaries.

Power Query Editor interface showing a table with columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, Profit Margin. The table is filtered by 'Table.Distinct(*Filtered Rows)'. The 'Order Details Summary' query is selected in the Queries list. The right pane shows the 'APPLIED STEPS' list, including 'Removed Duplicates'.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin
B-25601	1,275.00	-1148	7	Furniture	Bookcases	
B-25601	66.00	-12	5	Clothing	Stole	
B-25601	8.00	-2	3	Clothing	Hankkerchief	
B-25601	80.00	-56	4	Electronics	Electronic Games	
B-25602	168.00	-111	2	Electronics	Phones	
B-25602	424.00	-272	5	Electronics	Phones	
B-25602	2,617.00	1151	4	Electronics	Phones	
B-25602	561.00	212	3	Clothing	Saree	
B-25602	119.00	-5	6	Clothing	Saree	
B-25603	1,355.00	-60	5	Clothing	Trousers	
B-25603	24.00	-30	1	Furniture	Chairs	
B-25603	193.00	-166	3	Clothing	Saree	
B-25603	180.00	5	3	Clothing	Trousers	
B-25603	116.00	16	4	Clothing	Stole	
B-25603	107.00	36	6	Clothing	Stole	
B-25603	12.00	1	2	Clothing	Hankkerchief	
B-25603	38.00	18	1	Clothing	Kurti	
B-25604	65.00	17	2	Clothing	T-shirt	
B-25604	157.00	5	9	Clothing	Saree	
B-25605	75.00	0	7	Clothing	Saree	
B-25606	87.00	4	2	Clothing	Shirt	
B-25607	50.00	15	4	Clothing	Leggings	
B-25608	1,364.00	-1864	5	Furniture	Tables	
B-25608	476.00	0	3	Furniture	Chairs	

Power Query Editor interface showing a table with columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, Profit Margin. The table is filtered by 'Table.Distinct(*Filtered Rows)'. A 'Group By' dialog box is open, showing the 'Basic' tab. The 'Order ID' column is selected for grouping, and the 'Count Rows' operation is chosen. The 'New column name' is 'Order Count'.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin
B-25601	1,275.00	-1148	7	Furniture	Bookcases	
B-25601	66.00	-12	5	Clothing	Stole	
B-25601	8.00	-2	3	Clothing	Hankkerchief	
B-25601	80.00	-56	4	Electronics	Electronic Games	
B-25602	168.00	-111	2	Electronics	Phones	
B-25602	424.00	-272	5	Electronics	Phones	
B-25602	2,617.00	1151	4	Electronics	Phones	
B-25602	561.00	212	3	Clothing	Saree	
B-25602	119.00	-5	6	Clothing	Saree	
B-25603	1,355.00	-60	5	Clothing	Trousers	
B-25603	24.00	-30	1	Furniture	Chairs	
B-25603	193.00	-166	3	Clothing	Saree	
B-25603	180.00	5	3	Clothing	Trousers	
B-25603	116.00	16	4	Clothing	Stole	
B-25603	107.00	36	6	Clothing	Stole	
B-25603	12.00	1	2	Clothing	Hankkerchief	
B-25603	38.00	18	1	Clothing	Kurti	
B-25604	65.00	17	2	Clothing	T-shirt	
B-25604	157.00	5	9	Clothing	Saree	
B-25605	75.00	0	7	Clothing	Saree	
B-25606	87.00	4	2	Clothing	Shirt	
B-25607	50.00	15	4	Clothing	Leggings	
B-25608	1,364.00	-1864	5	Furniture	Tables	
B-25608	476.00	0	3	Furniture	Chairs	

Power Query Editor interface showing a query named "Table.Group(#\"Removed Duplicates\", {\"Order ID\", {\"Order Count\", each Table.RowCount(_), Int64.Type}})". The query results are displayed in a table with columns: Order ID, Order Count, and Profit Margin. The table contains 25 rows of data, including Order IDs and their corresponding counts.

Query Settings:

- PROPERTIES: Name: Order Details Summary
- APPLIED STEPS: Source, Promoted Headers, Changed Type, Added Custom, Changed Type1, Added Conditional Column, Filtered Rows, Removed Duplicates, Grouped Rows

2 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

Power Query Editor interface showing a query named "Table.Distinct(#\"Filtered Rows\")". The query results are displayed in a table with columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, and Profit Margin. The table contains 25 rows of data, including Order IDs, amounts, profits, quantities, categories, and sub-categories.

Query Settings:

- PROPERTIES: Name: Avg Profit by Category
- APPLIED STEPS: Source, Promoted Headers, Changed Type, Added Custom, Changed Type1, Added Conditional Column, Filtered Rows, Removed Duplicates

8 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Advanced Editor

Manage

Choose Columns

Remove Columns

Keep Rows

Remove Rows

Split Column

Group By

Data Type: Text

Use First Row as Headers

Replace Values

Merge Queries

Append Queries

Combine Files

Combine

Close

New Query

Data Sources

Parameters

Output Data

Query

Manage Columns

Reduce Rows

Sort

Transform

Queries [5]

Orders Data

Order Details

Sales target

Order Details Summary

Avg Profit by Category

fx

= Table.Group(#"Removed Duplicates", ("Category"), ({ "Average Profit", each List.Average([Profit]), type nullable number})))

Category

Average Profit

Valid100%

Error0%

Empty0%

Valid100%

Error0%

Empty0%

1 Furniture9.456790123

2 Clothing11.76290832

3 Electronics34.07142857

Query Settings

PROPERTIES

Name

Avg Profit by Category

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Added Custom

Changed Type1

Added Conditional Column

Filtered Rows

Removed Duplicates

Grouped Rows

2 COLUMNS, 3 ROWS

Column profiling based on top 1000 rows

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Table: Table.Group(*Removed Duplicates*, {"Month of Order Date"}, {"Total Target", each List.Sum([Target]), type nullable number})

Month of Order Date	Total Target
1 Apr-18	31400
2 May-18	31500
3 Jun-18	31600
4 Jul-18	31800
5 Aug-18	31900
6 Sep-18	34000
7 Oct-18	36100
8 Nov-18	36300
9 Dec-18	36400
10 Jan-19	43500
11 Feb-19	43600
12 Mar-19	43800

Data Modeling:

Established data model relationships between List of Orders and Order Details using Order ID, and between Order Details and Sales Target using Category. Verified and managed relationships using the Manage Relationships feature to ensure all relationships were active and correctly configured for accurate reporting and analysis.

New relationship

Select tables and columns that are related.

From table
List of Orders

CustomerName	Location	Order Date	Order ID
Hitika	Indore, Madh...	Monday, April...	B-25627
Ashmi	Indore, Madh...	Thursday, Apr...	B-25637
Yaanvi	Indore, Madh...	Tuesday, May...	B-25645

To table
Order Details

Category	Order ID	Profit	Profit Margin	Profit Status	Quantity	Sub-Cat
Clothing	B-25603	1	0.0833333333...	Profit	2	Hankerc...
Clothing	B-25608	23	0.0894941634...	Profit	5	Hankerc...
Clothing	B-25615	20	0.2941176470...	Profit	5	Hankerc...

Cardinality
One to many (1:*)

Cross-filter direction
Single

☒ Make this relationship active

☐ Assume referential integrity

☐ Apply security filter in both directions

Save **Cancel**

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Clipboard Get data Excel OneLake SQL Server Enter data Datasource Recent sources Transform Refresh data Queries Manage relationships Relationships New measure column Calculations New table Calculation group New parameter Parameters Manage roles View as Q&A setup Language Linguistic schema Sensitivity Publish

Order Details

- Σ Amount
- Category
- Order ID
- Σ Profit
- Σ Profit Margin
- Profit Status
- Σ Quantity
- Sub-Category
- Collapse ^

Avg Profit by Category

- Σ Average Profit
- Category
- Collapse ^

List of Orders

- CustomerName
- Location
- Order Date
- Order ID
- Collapse ^

Orders Data

- Σ Amount
- Category
- CustomerName
- Location
- Order Date
- Order ID
- Σ Profit
- Σ Quantity
- Collapse ^

Properties

Relationship

Table: Order Details Column: Order ID

Cardinality: Many to one (*:1)

Table: List of Orders Column: Order ID

Make this relationship active: Yes

Cross-filter direction: Single

Apply security filter in both directions: No

Apply changes Open relationship editor

Data

Tables Model

View and organize all of the items in your semantic model. [Learn how.](#)

Semantic model

- Calculation groups (0)
- Cultures (1)
- Expressions (0)
- Measures (0)
- Perspectives (0)
- Relationships (2)
 - Order Details[Order ID] <-- List o...
 - Order Details[Sub-Category] <-- ...
- Roles (0)
- Tables (8)
 - Avg Profit by Category
 - List of Orders
 - Order Details
 - Order Details Summary
 - Orders Data
 - Sales target
 - Sales Target Summary

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Clipboard Get data Excel OneLake SQL Server Enter data Datasource Recent sources Transform Refresh data Queries Manage relationships Relationships New measure column Calculations New table Calculation group New parameter Parameters Manage roles View as Q&A setup Language Linguistic schema Sensitivity Publish

Order Details

- Σ Amount
- Category
- Order ID
- Σ Profit
- Σ Profit Margin
- Profit Status
- Σ Quantity
- Sub-Category
- Collapse ^

Avg Profit by Category

- Average Profit
- Category
- Collapse ^

Orders Data

- Σ Amount
- Category
- CustomerName
- Location
- Order Date
- Order ID
- Collapse ^

Sales target

- Category
- Month of Order Date
- Σ Target
- Collapse ^

Properties

Relationship

Table: Order Details Column: Category

Cardinality: Many to one (*:1)

Table: Avg Profit by C... Column: Category

Make this relationship active: Yes

Cross-filter direction: Single

Apply security filter in both directions: No

Apply changes Open relationship editor

Data

Tables Model

View and organize all of the items in your semantic model. [Learn how.](#)

Semantic model

- Calculation groups (0)
- Cultures (1)
- Expressions (0)
- Measures (0)
- Perspectives (0)
- Relationships (1)
 - Order Details[Ca...
- Roles (0)
- Tables (8)
 - Avg Profit by Ca
 - List of Orders
 - Order Details
 - Order Details Su
 - Orders Data
 - Sales target
 - Sales Target Sun
- Columns (2)

Data Modeling Logic

Relationships were established between the tables to enable accurate reporting and filtering across datasets. The List of Orders table was connected with the Order Details table using Order ID, while the Order Details table was connected with the Sales Target table using Category.

The relationships were configured and verified using the Manage Relationships feature in Power BI to ensure they were active and correctly defined. Proper cardinality and cross-filter directions were selected based on business logic to maintain data integrity and support efficient analysis across related tables.

